Sean Nugroho

Software engineer & researcher

Full-stack engineering, high-performance computing, and selected projects from Indonesia to Japan. I build software, and explore what makes it work better.

seannugroho8@gmail.com github.com/Leaguemen linkedin.com/in/venantius-sean-ardi-nugroho

EXPERIENCE

P.T. Doval Sejahtera Investa Mar 2026 — Present
Full-stack engineer
Building internal web applications for tracking employee man-hours and managing invoice reimbursement, across frontend and backend.

P.T. Teleakses Solusindo Jun — Aug 2025
Software developer intern
A Go audit-trail backend handling over 3,000 requests per second, a Flutter frontend, and a Vue.js migration.

Saint Theodore Church Jul — Aug 2024
Full-stack web developer
End-to-end development and deployment of a church event application.

SELECTED WORK

Finos 2026
Full-stack development · Next.js
A workspace for work logs and reimbursements, built at P.T. Doval Sejahtera Investa. Sign-in required to use the workspace.

Pemilu HMIF 2025
Full-stack development · Next.js, T3 Stack, Vercel
An election platform for the STEI-K ITB student community. Led a team of 10 developers and built the voting API.

Pope's Mass web application 2024
Full-stack development · React, Node.js, MySQL
An information hub for Saint Theodore Church members attending the Pope's Mass. Independently built, deployed, and maintained.

Network security with machine learning 2024
Machine learning · Python, scikit-learn
K-Nearest Neighbor, Naive Bayes, and Decision Tree models built from scratch to detect network attacks, then compared with scikit-learn implementations.

MarioDB 2024
Database systems · Python
A database management system built from scratch in Python as a university group project, exploring what happens underneath a query.

Linux-based operating system 2024
Systems programming · C, Assembly
A command-line operating system built from scratch with a university team, using C and Assembly.

RESEARCH & PUBLICATION

Research in Japan, 2025–2026: studied how inference-time reasoning affects LLM-generated parallel code at the University of Electro-Communications, Tokyo.

ISC High Performance 2026

Impact of Inference-Time Reasoning on LLM-Based Parallel Code Generation: a Case-Study of Kokkos

Venantius Sean Ardi Nugroho, Shinobu Miwa · University of Electro-Communications, Tokyo · March 2026

Using 60 ParEval problems, we compared correctness, token cost, and CPU speedup across 4,800 GPT-5.1-generated Kokkos programs at four reasoning levels. Higher reasoning effort improved correctness but did not consistently improve parallel performance. With automated validation, sampling multiple no-effort solutions was more token-efficient; the analysis does not include human review costs.

EDUCATION

Bandung Institute of Technology (ITB)

2022 — present

B.Sc. in Informatics

Expected graduation: December 2026.

University of Electro-Communications, Tokyo

2025 — 2026

Research student · MEXT scholarship

SKILLS

Languages	Go, Python, Java, C/C++, TypeScript, SQL
Frameworks	React, Next.js, Vue.js, Flutter, Node.js, Fiber
Tools	PostgreSQL, MySQL, Git, Docker, Linux
Spoken	Indonesian (native) · English (IELTS 8.0) · Japanese (JLPT N3)